

REZA KARBASI

Machine Learning Engineer

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EDUCATION

University of Tehran

08/2019 - 01/2022

MS in Artificial Intelligence and Robotics; GPA: 18.86/20

Tehran, Iran

- Thesis: [An earthquake early warning system using smartphones as sensors](#)
- Supervisors: [Prof. Hadi Moradi](#), [Dr. Mahmoud Reza Hashemi](#), [Dr. Ali Moradi](#)
- Notable Courses: Deep Neural Network, Reinforcement Learning, Advanced Robotics

K. N. Toosi University of Technology

08/2014 - 01/2019

BS in Electrical Engineering; GPA: 17.4/20

Tehran, Iran

- Thesis: Fault detection in gas turbine
- Supervisor : [Dr. Mahdi Aliyari](#)
- Notable Courses: Pattern Recognition, Advanced Programming, Digital Control, Signal and Systems

PUBLICATION

InProgress **Karbasi, R.**, Rahimi, M., Vahabi, H. (2024). ECG paper record digitization to detect different peaks of the signal.

SKILLS

Languages English(Toefl score: 99/120), Persian: Native

Programming Python(Advanced), C(Advanced), C++(Intermediate), Matlab(Intermediate)

Frameworks Pytorch(Advanced), Tensorflow(Advanced), MLflow (Intermediate), ROS2 (Familiar), Simulink(Familiar)

Libraries Numpy, Pandas, Matplotlib, OpenCV

Hardware Keil, Stm32Cube, ARM, XMega, Arduino, Altium, Soldering

Misc Git, Linux, Docker, Software Testing, GTest, PyTest, Bash Scripting, REST API

EXPERIENCE

Machine Learning Engineer

03/2024 - Now

PresentID

Remote, France

- Implemented car tracking in surveillance cameras, fine-tuned **Ultralytics** to accurately identify different lanes and vehicles.
- Enhanced an OCR system using **EasyOCR** to detect and scan official documents. Integrated **pytest** for robust testing, utilized **MLflow** for experiment tracking, and synthesized datasets for model fine-tuning.
- Deployed a microservice for voice denoising using [rnnoise](#) within a Flask application.

Automotive Software Engineer

04/2023 - 03/2024

SoftwareMotion

Remote, China

- Made a **CI/CD** pipeline in **Gitlab** with **GTest** for the Autosar-based codebase, reducing on-field testing through automated unit and integration tests by 30%
- Built a **PyQT** visualization software for AEB and ACC systems, integrating BEV plots to drastically reduce debugging time. Processed MF4 and CAN signals (BLF, DBC) to expedite vehicle issue diagnosis.
- Evaluated and integrated advanced path planning solutions, such as lattice-based and [Apollo's](#) EM planner. Also, developed obstacle avoidance algorithms with **DP** and **A*** techniques, validated through a custom Python-based simulator.

Machine Learning Engineer

03/2022 - 03/2023

[Hara AI Company](#)

Tehran, Iran

- Automated audio file categorization by gender and age using Google's [YamNet](#) (in the TensorFlow framework) as the backbone, fine-tuning the model with the company's labeled data. Developed a custom metric to accurately evaluate classification accuracy, with equal emphasis on precision and recall.
- Conducted an in-depth assessment and visualization of the company's Speech-to-Text (**STT**) models under varied real-world conditions (such as noise and reverberation) to pinpoint vulnerabilities.

AI Engineer

10/2022 - 12/2023

[Iran's National Elites Foundation](#)

Tehran, Iran

Led a team to develop a microservice-based vehicle monitoring platform, focusing on **graph-based path planning**. Enhanced navigation precision and responsiveness using **Dynamic Programming** and **A***, achieving a tenfold increase in processing speed.

Instructor

06/2021 - 09/2021

[Aras Academy](#)

Tehran, Iran

Presented **an Introduction to Reinforcement Learning** at Aras Academy's 2021 summer school, basing the syllabus on Sutton's book, "Reinforcement Learning: An Introduction."

Embedded System Engineer

12/2017 - 06/2019

[Raiwan](#)

Tehran, Iran

- Developed **Inner Hospital Communication** boards using **STM32 F1** microcontrollers, enabling efficient nurse and patient calls, deployed in 2+ hospitals.
- Developed **Inner Hospital Communication** boards with **STM32 F1** microcontrollers, incorporating audio signal transportation via **RS485** protocol, and implemented audio recording and playback on digital boards. Enabled efficient real-time communication between nurses and patients, successfully deployed in over 2 hospitals.
- Designed **Taxi Payment Device** frame via CNC and worked on the device's **ESP32** integration.

Robotic Researcher

05/2015 - 09/2017

[KN2C Robotic Lab](#)

Tehran, Iran

- Led KN2C's SSL electronics team, improving robot performance by designing, soldering, and maintaining Xmega and ST processor boards for motor control and **PID** calibration. Collaborated with mechanical and software teams to integrate system enhancements.
- Boosted ball detection precision by implementing a **Kalman Filter** in **C++**, leveraging the Armadillo library.
- Crafted a real-time GUI in **C#** using Visual Studio for simultaneous condition monitoring of all robots, enhancing strategic decision-making.

Teaching Assistant

served as a Teaching Assistant in some courses:

- **Deep Neural Networks**: Spring 2022, working with [Dr. Reshad Hosseini](#)
- **Reinforcement Learning**: Fall 2021, working with [Prof. Majid Nili Ahmadabadi](#)
- **System Identification**: Spring 2020, working with [Prof. Babak Nadjar Araabi](#)
- **Pattern Recognition**: Fall 2022, Fall 2019, Fall 2017, working with [Prof. Hamid Abrishami Moghaddam](#)

NOTABLE PROJECTS

- Implemented an Adaptive Neuro-Fuzzy Inference System (**ANFIS**) algorithm in **Pytorch** for EEG signal classification, facilitating video game control through brain activity.
- Developed a Metatrader trading bot using a technical strategy, enhanced by a **Deep Q Network** in **PyTorch**. Leveraged a socket connection for real-time data and analysis of historical data from 2007-2019 in Metatrader. Documented a version on this [Medium blog](#) and this [Github repo](#).
- Dockerized a **face recognition system** in videos using **Flask** and **deepface** library for identifying individuals by comparing faces with reference images, handling video processing, face embedding, matching, and output generation for applications like surveillance. [Project link](#).
- Implemented the encoder component of a **Transformer** network for a translation application (English to Persian) as part of a project in the Deep Learning course.
- Applied an image captioning application using **Resnet18** and **LSTM** models in **PyTorch**.
- Led the web crawling of the [Soroush app](#) for the Big Data Course's final project, utilizing **Selenium** for data extraction and **Kafka** for transmission. Facilitated team collaboration for data analysis, with all contributions and code housed in our [GitHub repository](#).
- Developed a gym environment with a Deep RL **A3C** agent to optimize station power settings and maximize coverage area while minimizing conflicts.
- Localized Anki robot in a **Gazebo** environment using **particle filter** and **EKF** algorithms.
- Led sentiment analysis of mobile phone reviews on Digikala for a Social Network course project, developing a metric to quantify device opinions. Spearheaded website crawling with **Selenium** and crafted the analysis network. Code available in [this repository](#).

HONORS & CERTIFICATES

Iran Open RoboCup: Ranked 5th in 2016's Small Sized League (SSL)

Snake League: Ranked 3rd in 2019's Snake League in Nasir Cup

The National University Entrance Exam for M.Sc. degree in AI: Ranked 28th(top 0.2%)

National University Entrance Exams in Mathematics: Ranked 888th(top 0.7%)

Certificate in Database Principles and SQL Server: [Link to the certificate](#)

Certificate in Django Web Framework: [Link to the certificate](#)

REFERENCES

Dr. Reshad Hosseini Assistant Professor, Department of Electrical and Computer Engineering, Cofounder of Hara AI company, UT, Tehran, Iran
Email: reshad.hosseini@ut.ac.ir

Dr. Mahdi Aliyari-Shoorehdeli Associate Professor, Control and Mechatronics Engineering, K. N. Toosi University of Technology, Tehran, Iran
Email: aliyari@kntu.ac.ir

Dr. Ali Moradi Associate Professor, Institute of Geophysics, UT, Tehran, Iran
Email: asmoradi@ut.ac.ir